LAB 05

1. Compress a file by gzip and bzip2 commands and decompress it again.
State the differences between two commands.

```
[mina_shafik@localhost ~]$ ls
list_dir  modify  mydir  mydtxt.txt  myfile  myteam  oldpasswd  snake  source_dir  source_file  target_dir  testdir  testfile
[mina_shafik@localhost ~]$ sudo gzip snake
[mina_shafik@localhost ~]$ ls
list_dir  modify  mydir  mydtxt.txt  myfile  myteam  oldpasswd  snake.gz  source_dir  source_file  target_dir  testdir  testfile
[mina_shafik@localhost ~]$ sudo gunzip snake
[mina_shafik@localhost ~]$ ls
list_dir  modify  mydir  mydtxt.txt  myfile  myteam  oldpasswd  snake  source_dir  source_file  target_dir  testdir  testfile
[mina_shafik@localhost ~]$
```

2. What is the command used to view the content of a compressed file.

sudo zcat filename.gz

3. Backup /etc directory using tar utility.

```
[mina_shafik@localhost etc]$ sudo tar -cf etc-backup.tar /etc
tar: Removing leading `/' from member names
tar: /etc/etc-backup.tar: file is the archive; not dumped
[mina_shafik@localhost etc]$ ls  /etc/
adjtime             csh.cshrc                GREP_COLORS  kernel          makedumpfile.conf.sample  pkcs11                  selinux         sysctl.d
aliases             csh.login                groff        krb5.conf       man_db.conf               pki                     services        systemd
alternatives        dbus-1                   group        krb5.conf.d     microcode_ctl             pm                      sestatus.conf   system-release
anacrontab          default                  group-       ld.so.cache     mime.types                popt.d                  shadow          system-release-cpe
audit               depmod.d                 grub2.cfg    ld.so.conf      mke2fs.conf               printcap                shadow-         terminfo
authselect          dhcp                     grub.d       ld.so.conf.d    modprobe.d                profile                 shells          tmpfiles.d
bash_completion.d   DIR_COLORS               gshadow      libaudit.conf   modules-load.d            profile.d               skel            tpm2-tss
bashrc              DIR_COLORS.lightbgcolor  gshadow-     libibverbs.d    motd                      protocols               ssh             udev
binfmt.d            dnf                      gss          libnl           motd.d                    rc.d                    ssl             vconsole.conf
chrony.conf         dracut.conf              host.conf    libreport       mtab                      rc.local                sssd            vimrc
chrony.keys         dracut.conf.d            hostname     libssh          NetworkManager            redhat-release          statetab.d      virc
cifs-utils          environment              hosts        libuser.conf    networks                  resolv.conf             subgid          wgetrc
cron.d              etc-backup.tar           httpd        locale.conf     nftables                  rocky-release           subgid-         X11
cron.daily          ethertypes               inittab      localtime       nsswitch.conf             rocky-release-upstream  subuid          xattr.conf
cron.deny           exports                  inputrc      login.defs      nsswitch.conf.bak         rpc                     subuid-         xdg
cron.hourly         filesystems              iproute2     logrotate.conf  openldap                  rpm                     sudo.conf       yum
cron.monthly        firewalld                issue        logrotate.d     opt                       rsyslog.conf            sudoers         yum.conf
crontab             fonts                    issue.d      lvm             os-release                rsyslog.d               sudoers.d       yum.repos.d
cron.weekly         fstab                    issue.net    machine-id      pam.d                     rwtab.d                 sudo-ldap.conf
crypto-policies     gcrypt                   kdump        magic           passwd                    sasl2                   sysconfig
crypttab            gnupg                    kdump.conf   mailcap         passwd-                   security                sysctl.conf
[mina_shafik@localhost etc]$
```

4. Starting from your home directory,
find all files that were modified in the last two day.

```
[mina_shafik@localhost etc]$ sudo find ~ -type f -mtime -2
/home/mina_shafik/.bash_history
/home/mina_shafik/etc-backup.tar
[mina_shafik@localhost etc]$
```

5. Starting from /etc, find files owned by root user.

```
[mina_shafik@localhost etc]$ sudo find /etc/ -user root
/etc/
/etc/mtab
/etc/fstab
/etc/crypttab
/etc/lvm
/etc/lvm/devices
/etc/lvm/devices/system.devices
/etc/lvm/archive
/etc/lvm/archive/rl_00000-899613452.vg
/etc/lvm/backup
/etc/lvm/backup/rl
/etc/lvm/cache
/etc/lvm/lvm.conf
```

6. Find all directories in your home directory.

```
[mina_shafik@localhost etc]$ sudo find ~ -type d
/home/mina_shafik
/home/mina_shafik/myteam
/home/mina_shafik/testdir
/home/mina_shafik/mydir
/home/mina_shafik/source_dir
/home/mina_shafik/target_dir
/home/mina_shafik/target_dir/source_dir
/home/mina_shafik/list_dir
[mina_shafik@localhost etc]$
```

7. Write a command to search for all files on the system that, its name is ".profile".

```
Last login: Sun Mar 15 17:36:59 2026
[root@localhost ~]# sudo find / -type f -name ".profile"
find: '/proc/62257': No such file or directory
[root@localhost ~]#
```

8. Identify the file types of the following:
/etc/passwd, /dev/pts/0, /etc, /dev/sda

```
[root@localhost ~]# file /etc/passwd /dev/pts/0 /etc /dev/sda
/etc/passwd: ASCII text
/dev/pts/0:  character special (136/0)
/etc:        directory
/dev/sda:    block special (8/0)
[root@localhost ~]#
```

9. List the inode numbers of /, /etc, /etc/hosts

```
[root@localhost ~]# ls -id / /etc /etc/hosts
33575041 drwxr-xr-x. 78 root root 8192 Mar 15 17:34 /etc
     128 dr-xr-xr-x. 25 root root 4096 Mar  7 01:09 /
33794713 -rw-r--r--.  1 root root  158 Jun 23  2020 /etc/hosts
[root@localhost ~]#
```

10.Copy /etc/passwd to your home directory, use command diff and Edit in the file you copied, and then use the command again, and check the output.

```
[mina@localhost ~]$ cp /etc/passwd ~/passwd_copy
[mina@localhost ~]$ diff /etc/passwd ~/passwd_copy
[mina@localhost ~]$ cat ~/passwd_copy
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
systemd-coredump:x:999:997:systemd Core Dumper:/:/sbin/nologin
dbus:x:81:81:System message bus:/:/sbin/nologin
sssd:x:998:995:User for sssd:/:/sbin/nologin
tss:x:59:59:Account used for TPM access:/dev/null:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
chrony:x:997:994:chrony system user:/var/lib/chrony:/sbin/nologin
systemd-oom:x:992:992:systemd Userspace OOM Killer:/:/usr/sbin/nologin
mina:x:1000:1000:mina shafik:/home/mina:/bin/bash
[mina@localhost ~]$
```

11.Create a symbolic link of /etc/passwd in /boot.

```
[mina@localhost ~]$ sudo ln -s /etc/passwd /boot/passwd_link
[mina@localhost ~]$ ls -l /boot/passwd_link
lrwxrwxrwx. 1 root root 11 Mar 17 22:23 /boot/passwd_link -> /etc/passwd
[mina@localhost ~]$
```

12.Create a hard link of /etc/passwd in /boot. Could you? Why?

Hard link → NOT possible (different filesystems)
Symbolic link → Works fine

```
[mina@localhost ~]$ df /etc/passwd /boot
Filesystem          1K-blocks    Used Available Use% Mounted on
/dev/mapper/rl-root  17811456 1216580  16594876   7% /
/dev/sda1             1038336  224968    813368  22% /boot
[mina@localhost ~]$ ln /etc/passwd /boot/passwd_hard
ln: failed to create hard link '/boot/passwd_hard' => '/etc/passwd': Invalid cross-device link
[mina@localhost ~]$ ln -s /etc/passwd /boot/passwd_link
ln: failed to create symbolic link '/boot/passwd_link': File exists
```

13.Add disk and create a mount point to empty directory with 2 ways temporary and permenant

```
[root@localhost ~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda           8:0    0   20G  0 disk
├sda1         8:1    0    1G  0 part /boot
└sda2         8:2    0   19G  0 part
  ├rl-root 253:0    0   17G  0 lvm  /
  └rl-swap 253:1    0    2G  0 lvm  [SWAP]
sr0          11:0    1  8.8G  0 rom
[root@localhost ~]# ^C
[root@localhost ~]# fdisk /dev/sdb

Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

fdisk: cannot open /dev/sdb: No such file or directory
[root@localhost ~]# mkfs.ext4 /dev/sdb1
mke2fs 1.46.5 (30-Dec-2021)
The file /dev/sdb1 does not exist and no size was specified.
[root@localhost ~]# mkdir /mnt/mydisk
mkdir: cannot create directory '/mnt/mydisk': File exists
[root@localhost ~]# mount /dev/sdb1 /mnt/mydisk
mount: /mnt/mydisk: special device /dev/sdb1 does not exist.
[root@localhost ~]# /etc/fstab
bash: /etc/fstab: Permission denied
```